

L-14 Chapter - 5.3, 5.4

LU decomposition

- More efficient than Gaussian Elimination
- Do not need to include bias/constant calculation all the time.
- Need upper & lower matrix.

Ex.

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$x + 4y + 9z = 16$$

Step-1

$$A^{(1)} = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$\leftarrow \text{for } a_{21} \Rightarrow R_2 = R_2 - \left(\frac{3}{2}\right)R_1 \\ = R_2 - 1.5R_1$$

$$\underline{F^{(1)}} = \begin{bmatrix} 1 & 0 & 0 \\ -1.5 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix}$$

$$\leftarrow \text{for } a_{31} \Rightarrow R_3 = R_3 - \left(\frac{1}{2}\right)R_1 \\ = R_3 - 0.5R_1$$

keep 0 at step(1).

Step-2
Now

$$A^{(2)} = F^{(1)} \cdot A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -1.5 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 3.5 & 8.5 \end{bmatrix}$$

$$\underline{F^{(2)}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix}$$

$$\leftarrow \text{for } a_{32} \Rightarrow R_3 = R_3 - \left(\frac{3.5}{0.5}\right)R_2 \\ \underline{[use A^2]} \Rightarrow R_3 = R_3 - 7R_2$$

$$A^{(3)} = [U] = F^{(2)} \cdot A^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 3.5 & 8.5 \end{bmatrix}$$

we got upper triangular $[U]$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 0 & -2 \end{bmatrix}$$

Lower triangular, $[L] = \begin{bmatrix} 1 & 0 & 0 \\ 1.5 & 1 & 0 \\ 0.5 & 7 & 1 \end{bmatrix}$

Make sure to change the signs!!

Now

$$[L] [TempMatrix] = [B]$$

3×3 3×1 3×1

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1.5 & 1 & 0 \\ 0.5 & 7 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$\boxed{a = 10}$$

$$1.5a + b + 0 = 18 \Rightarrow \boxed{b = 3}$$

$$0.5a + 7b + c = 16 \Rightarrow \boxed{c = -10}$$

Now Final formula:

$$[U][variables] = [TempMatrix]$$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 0 & -2 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ -10 \end{bmatrix}$$

use bottom to top: Solve $\longrightarrow x = 7, y = 9, z = 5$

Another Example

Given $2x + 4y + 2z = 10$

$$4x - 2y + z = 15$$

$$8x + 4y + 3z = 20$$

$$A^{(1)} = \begin{bmatrix} 2 & 4 & 2 \\ 4 & -2 & 1 \\ 8 & 4 & 3 \end{bmatrix}$$

B

Step 1:

$$F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

for $a_{21} \Rightarrow$

$$R_2 = R_2 - \left(\frac{4}{2}\right)R_1$$

$$= R_2 - 2R_1$$

for $a_{31} \Rightarrow R_3 = R_3 - \left(\frac{8}{2}\right)R_1$

$$= R_3 - 4R_1$$

Step-2
Now

$$A^{(2)} = F^{(1)} \cdot A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 & 2 \\ 4 & -2 & 1 \\ 8 & 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & 2 \\ 0 & -10 & -3 \\ 0 & -12 & -5 \end{bmatrix}$$

$$F^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1.2 & 1 \end{bmatrix}$$

use $A^{(2)}$

for $a_{32} \Rightarrow R_3 = R_3 - \left(\frac{-12}{-10}\right)R_2$

$$= R_3 - 1.2R_2$$

$$[U] = A^{(3)} = F^{(2)} \cdot A^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1.2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 & 2 \\ 0 & -10 & -3 \\ 0 & -12 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & 2 \\ 0 & -10 & -3 \\ 0 & 0 & -1.4 \end{bmatrix}$$

So,

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1.2 & 1 \end{bmatrix}$$

Note!

Make sure to
change signs!

Now

$$[L] \cdot [\text{Temp Matrix}] = [B]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1.2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 20 \end{bmatrix}$$

$$a = 10$$

$$2a + b + 0 = 15 \Rightarrow b = -5$$

$$4a + 1.2b + c = 20 \Rightarrow c = -14$$

Finally!

$$[U] \cdot [\text{variables}] = [\text{Temp Matrix}]$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 0 & -10 & -3 \\ 0 & 0 & -14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \\ -14 \end{bmatrix}$$

use: Bottom to top $\uparrow$

$$-14z = -14 \Rightarrow z = 10$$

$$-10y - 3z = -5 \Rightarrow y = -2.5$$

$$2x + 4y + 2z = 10 \Rightarrow x = 0$$

$$\therefore x = 0, y = -2.5, z = 10$$


Pivoting

- When any $x/y/z$ is missing, then it's no longer Square Matrix.
- Here, if there is 0 in diagonal then, we can't use gaussian / LU decomposition.
- In that case, need Pivoting.

Pivoting : Swapping any row/column.

$$\begin{bmatrix} 0 & 5 & 7 \\ 9 & 2 & 6 \\ 7 & 8 & 1 \end{bmatrix} \xrightarrow{\text{Row Swap}} \begin{bmatrix} 9 & 2 & 6 \\ 0 & 5 & 7 \\ 7 & 8 & 1 \end{bmatrix}$$

↳ you can also swap column

$$\begin{bmatrix} 5 & 0 & 7 \\ 2 & 9 & 6 \\ 8 & 7 & 1 \end{bmatrix}$$

Why pivoting??

$$\text{for } a_{21} \Rightarrow R_2 - \left(\frac{9}{0}\right) R_1$$

∞

so, no 0 in diagonals.